

Research Statement

Jacob (Jake) Carlson*

September 2026

I am an econometrician broadly interested in developing statistical and causal inference methods that leverage unstructured data sources—e.g., text, image, audio, video, location trace, or multi-modal data. My work extends methods in semiparametric and high-dimensional statistics as well as in AI and machine learning to deliver principled, general-purpose frameworks for econometric analysis with unstructured data. The applications of such work span many fields, but are especially relevant in political economics and development economics—fields in which unstructured data are often critical for forming novel economic insights.

My graduate studies have focused on several distinct projects in support of this broad research focus, including: (1) my job market paper, which develops a new framework for making statistically and scientifically principled “discoveries” from unstructured datasets, accomplished in part by developing new tools for high-dimensional multiple testing; (2) a project that builds on insights from semiparametric inference for missing data to inform a general framework for robust and efficient inference with AI predictions based on unstructured data [Carlson and Dell, 2025]; (3) a project that develops a fully nonparametrically founded framework for dynamic causal inference with a single time series (the “potential system”), which directly motivates a role for flexibly modeled functions of unstructured data in time series causal inference [Carlson and Shephard, 2026]; and (4) a variety of projects that directly focus on the large-scale curation of new unstructured data sources, made possible with new innovations in AI/ML methods [Dell et al., 2023, Carlson et al., 2024a, Bryan et al., 2023, Shen et al., 2021]. I have also worked on econometrics projects that treat topics in misspecification and clustered standard errors, respectively [Andrews et al., 2026, Carlson et al., 2024b].

My job market paper, “Interpretable Discovery from Unstructured Data: A High-Dimensional Approach,” proposes an automatic, general-purpose framework for making discoveries from unstructured data: it leverages recent methods from the literature on AI interpretability to transform unstructured datasets into high-dimensional, structured datasets of interpretable concept measurements; specifies concept-level parameters and null hypotheses based on this transformed dataset; tests these hypotheses using algorithms validated by new results in high-dimensional multiple testing, producing a selected set (“discoveries”); and both generates and evaluates human-interpretable natural language descriptions of these discoveries. The proposed framework has few researcher degrees of freedom, is robust to data snooping, mitigates under-exploration, and facilitates fast and inexpensive sensitivity analysis and replication. The framework is made possible by extending two frontier econometric frameworks: the high-dimensional “many approximate means” framework of Belloni et al. [2018] and the generalized FWER controlling framework of Romano and Wolf [2007]. The framework also provides a novel statistical formalization of the “detection scoring” AI interpretability framework of Paulo et al. [2024], showcasing how econometric insights may be readily applied to AI evaluation pipelines.

*Email: jacob_carlson@g.harvard.edu

The framework is applied to a variety of both descriptive and causal analyses in prominent empirical economics papers, recovering original discoveries and making new discoveries too.

Sometimes a researcher is not interested in unsupervised discovery from unstructured data, but instead in leveraging predictions of known quantities of interest (e.g., based on theory or previous empirical studies) in downstream econometric analyses. In joint work with Melissa Dell, “A Unifying Framework for Robust and Efficient Inference with Unstructured Data” [Carlson and Dell, 2025], we propose a simple perspective and econometric framework for robust and efficient inference in such a setting: that predictions from unstructured data using deep neural networks can be viewed as imputations of low-dimensional missing structured data, and, as such, statistical methods for missing data (including many of those for causal inference) can be applied under appropriate adaptation. With this insight, we develop doubly robust and semiparametrically efficient estimators of parameters defined by moment restrictions using AI predictions from unstructured data, nesting the results of other popular frameworks along the way (e.g., Angelopoulos et al. [2023]), and further generalize to settings where predictions from unstructured data are aggregated before being analyzed in downstream linear regressions.

The semi/nonparametric identification results for causal parameters that motivate AI/ML predictors in cross-sectional studies are largely absent in studies of pure (single) time series. This is because most modern efforts in causal inference for a single time series (such as in macroeconometrics) are parametrically founded or projection focused. Other dynamic nonparametric causal inference efforts exist, but largely focus on panel data, and estimands of interest typically differ from the single time series setting. To bridge this gap in the literature, in joint work with Neil Shephard, “When are time series predictions causal? The potential system and dynamic causal effects” [Carlson and Shephard, 2026], we develop a nonparametric, potential outcomes-style foundational model for dynamic causal inference with a single time series: a time series system we call the “potential system.” Using the potential system, we derive novel but familiar looking estimands and nonparametric identification results for dynamic causal parameters in a single time series, and thereby motivate time series applications of causal machine learning methods long used in the cross-sectional causal inference literature. Such results, in turn, directly suggest how unstructured data may be leveraged in identification and estimation of dynamic causal effects.

Pure AI/ML methods development for curating new sources of unstructured data was the subject of several projects at the beginning of my graduate studies. In “Efficient OCR for Building a Diverse Digital History” [Carlson et al., 2024a], joint work with Melissa Dell and Tom Bryan (and an ACL *oral presentation*), we develop a new neural architecture for performing optical character recognition (OCR) in low-resource languages, based on the insight that, in the interest of sample efficiency, OCR can be treated as an information retrieval problem in a particular metric space. This OCR method was later adopted in “American Stories: A Large-Scale Structured Text Dataset of Historical U.S. Newspapers” [Dell et al., 2023], helping enable the digitization of over 20 million scans in the Library of Congress’s public domain “Chronicling America” collection.

I plan to continue centering econometric methods for unstructured data in my research agenda, soon expanding into work that considers unstructured data as treatment: both simplifying causal inference based on latent treatments exposed via unstructured data, and optimizing generative AI to deliver unstructured (text or image) treatments with quantifiably large causal effects. I am further interested in extending the frameworks I have already developed to address longstanding questions in macroeconomics (e.g., on monetary policy causal effects) as well as nascent questions in AI evaluation (e.g., on measurement error in AI usage data classification).

References

- Jacob Carlson and Melissa Dell. A Unifying Framework for Robust and Efficient Inference with Unstructured Data, 2025. URL <https://arxiv.org/abs/2505.00282>. Version Number: 2.
- Jacob Carlson and Neil Shephard. When are time series predictions causal? The potential system and dynamic causal effects, 2026. URL <https://arxiv.org/abs/2603.20394>. Version Number: 1.
- Melissa Dell, Jacob Carlson, Tom Bryan, Emily Silcock, Abhishek Arora, Zejiang Shen, Luca D’ Amico-Wong, Quan Le, Pablo Querubin, and Leander Heldring. American Stories: A Large-Scale Structured Text Dataset of Historical U.S. Newspapers. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 80744–80772. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/ffeb860479ccae44d84c0de32acd693d-Paper-Datasets_and_Benchmarks.pdf.
- Jacob Carlson, Tom Bryan, and Melissa Dell. Efficient OCR for Building a Diverse Digital History. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8105–8115, Bangkok, Thailand, 2024a. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.440. URL <https://aclanthology.org/2024.acl-long.440>.
- Tom Bryan, Jacob Carlson, Abhishek Arora, and Melissa Dell. EfficientOCR: An Extensible, Open-Source Package for Efficiently Digitizing World Knowledge. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 579–596, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-demo.52. URL <https://aclanthology.org/2023.emnlp-demo.52>.
- Zejiang Shen, Ruochen Zhang, Melissa Dell, Benjamin Charles Germain Lee, Jacob Carlson, and Weining Li. LayoutParser: A Unified Toolkit for Deep Learning Based Document Image Analysis. In Josep Lladós, Daniel Lopresti, and Seiichi Uchida, editors, *Document Analysis and Recognition – ICDAR 2021*, volume 12821, pages 131–146. Springer International Publishing, Cham, 2021. ISBN 978-3-030-86548-1 978-3-030-86549-8. doi: 10.1007/978-3-030-86549-8_9. URL https://link.springer.com/10.1007/978-3-030-86549-8_9. Series Title: Lecture Notes in Computer Science.
- Isaiah Andrews, Harvey Barnhard, and Jacob Carlson. True and Pseudo-True Parameters, April 2026. URL <http://arxiv.org/abs/2604.15563>. arXiv:2604.15563 [econ.EM].
- Jacob Carlson, Trevor Incerti, and P. M. Aronow. Dyadic Clustering in International Relations. *Political Analysis*, 32(2):186–198, April 2024b. ISSN 1047-1987, 1476-4989. doi: 10.1017/pan.2023.26. URL https://www.cambridge.org/core/product/identifier/S1047198723000268/type/journal_article.
- Alexandre Belloni, Victor Chernozhukov, Denis Chetverikov, Christian Hansen, and Kengo Kato. High-Dimensional Econometrics and Regularized GMM, 2018. URL <https://arxiv.org/abs/1806.01888>. Version Number: 2.

Joseph P. Romano and Michael Wolf. Control of generalized error rates in multiple testing. *The Annals of Statistics*, 35(4), August 2007. ISSN 0090-5364. doi: 10.1214/009053606000001622. URL <https://projecteuclid.org/journals/annals-of-statistics/volume-35/issue-4/Control-of-generalized-error-rates-in-multiple-testing/10.1214/009053606000001622.full>.

Gonçalo Paulo, Alex Mallen, Caden Juang, and Nora Belrose. Automatically Interpreting Millions of Features in Large Language Models, 2024. URL <https://arxiv.org/abs/2410.13928>. Version Number: 3.

Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. Prediction-powered inference. *Science*, 382(6671):669–674, November 2023. doi: 10.1126/science.adi6000. URL <https://www.science.org/doi/10.1126/science.adi6000>. Publisher: American Association for the Advancement of Science.